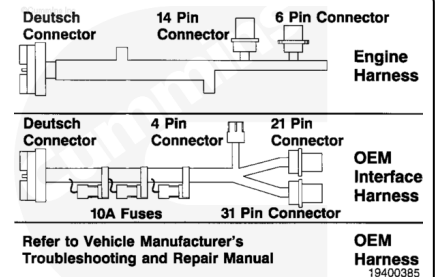


Trainings ▾

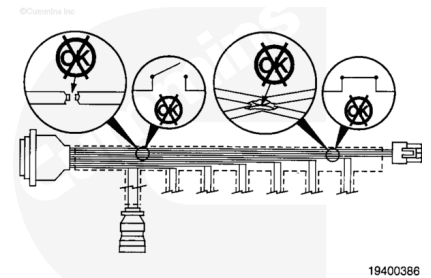
General Information

QSK-equipped engines use three separate wiring harnesses to control the engine and some of the vehicle operations.

1. Engine harness
2. OEM interface harness
3. OEM harness.



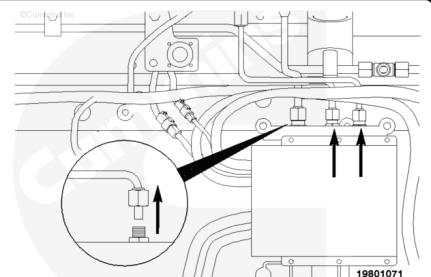
Replace a harness if there is an open circuit or a short circuit found under the protective covering of the harness body.



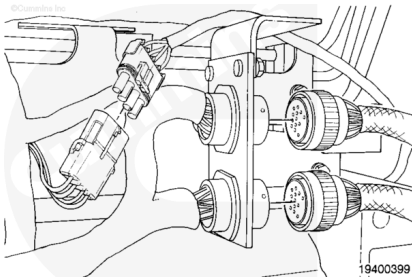
Remove

QSK19 Engines

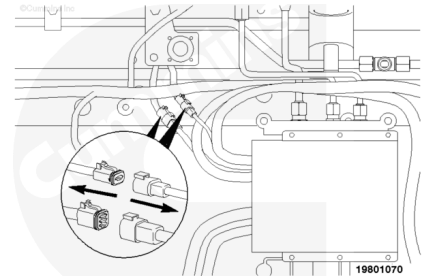
Remove the fuel supply tubes from the top of the control valve body to gain access to the OEM interface harness. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/19/19-006-024.html>) in the Troubleshooting and Repair Manual, QSK19 Series Engines, Bulletin 3666098 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666098.html>).



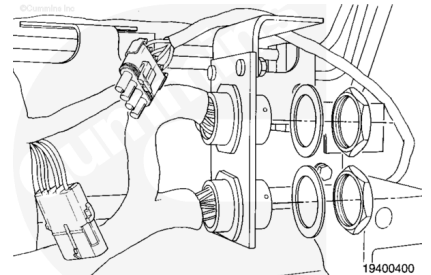
Disconnect the 21- and 31-pin connectors from the OEM harness.



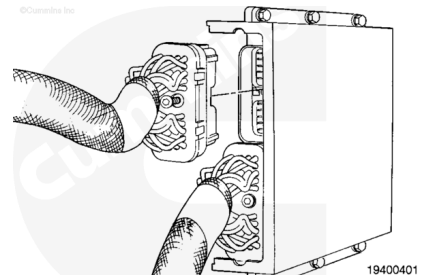
Disconnect the 2- and 6-pin Deutsch connectors.



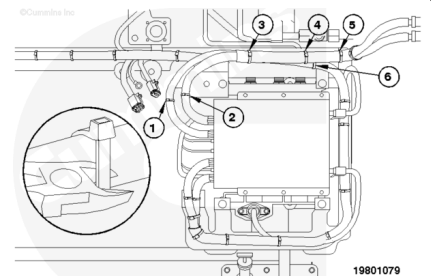
Remove the 21- and 31-pin connectors from the harness support brackets.



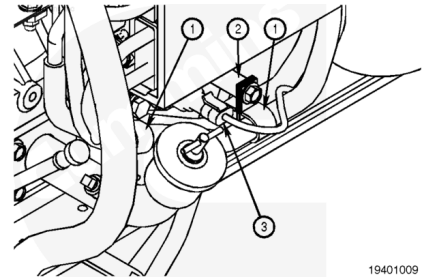
Use a 4-mm [5/32-in] hexagon head wrench to disconnect the OEM interface harness Deutsch connector from the ECM.



Remove the six wire ties from the harness support bracket and remove the OEM interface harness from the engine.

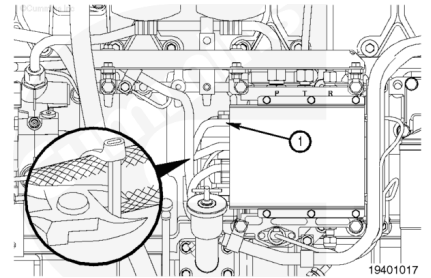


Remove the p-clips (1) and bracket (2) near the bottom of the ECM near the ambient air pressure sensor (3).



Remove the eight wire ties securing the OEM harness to the engine harness.

Disconnect connector B (1) from the ECM.



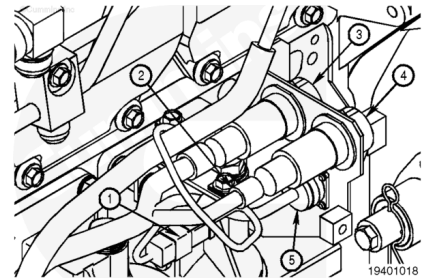
Disconnect the datalink connector (1) connecting the OEM harness to the engine harness.

Remove the p-clips (2) near the lower right corner of the number 5 cam follower cover.

Remove the connector nuts securing the 21- and 31-pin connectors (3, 4) to the bracket.

Remove the four capscrews (5) securing the 9-pin Deutsch connector to the bracket.

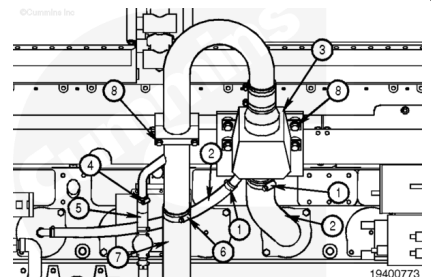
Remove the OEM harness.



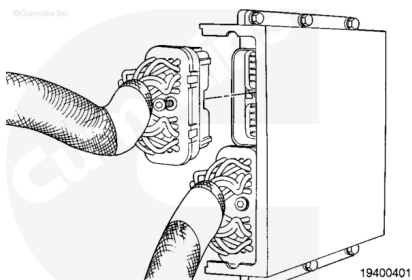
QSK45 and QSK60

Remove the breather box assembly.

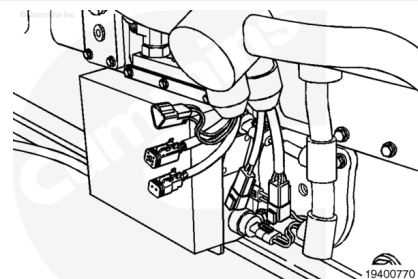
- Loosen the two hose clamps (1) and disconnect the two hoses (2) on the bottom of the breather box (3).
- Loosen the hose clamp (4) and disconnect the blowby pressure sensor hose (5).
- Loosen the hose clamp (6) and disconnect the lower breather hose (7).
- Remove the six attaching bolts (8) and the breather box assembly.



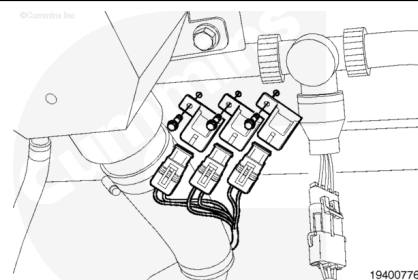
Use a 4-mm [5/32-in] hexagon head wrench to disconnect the OEM interface harness Deutsch connector from the ECM.



Disconnect the three datalink connectors.

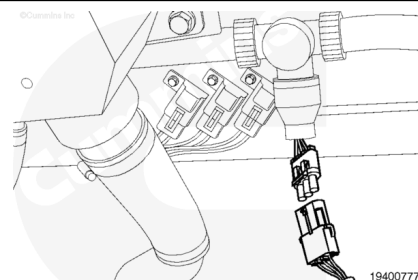


Remove the screws attaching the three fuse holders from the engine wiring harness bracket.

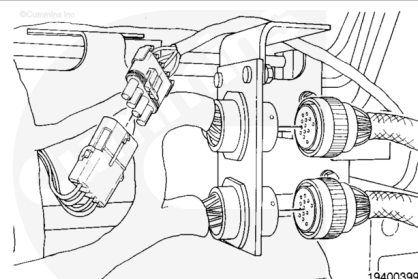


If CENSE™-equipped, disconnect the OEM interface harness from the CENSE™ harness.

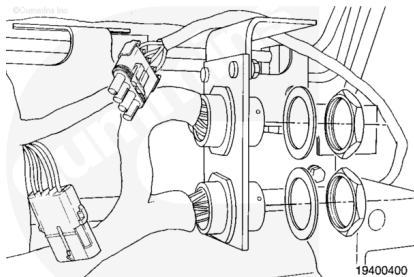
Note : On engines not equipped with CENSE™, the CENSE™ connectors will be connected to each other and must remain that way for proper operation.



Disconnect the 21- and 31-pin connectors from the OEM harness.



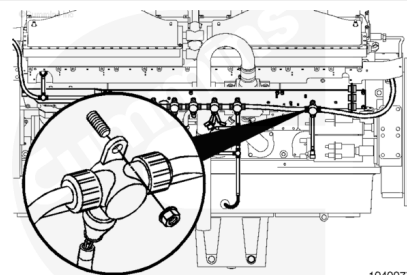
Remove the 21- and 31-pin OEM interface harness connectors from the engine harness support bracket.



Remove the six nuts holding the three wire clamps that secure the OEM interface harness.

Remove the OEM interface harness from the engine.

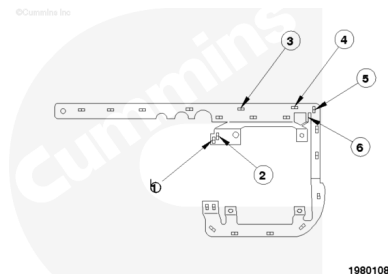
Separate and remove the wire clamps from the OEM interface harness.



Install

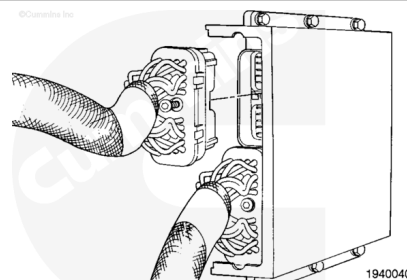
QSK19 Engines

Feed six wire ties through the clips of the harness support bracket.

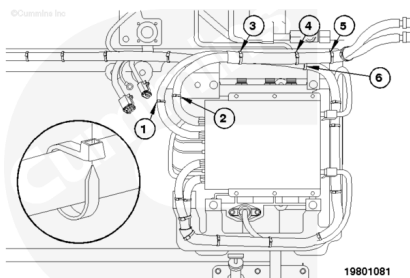


Connect the OEM interface harness Deutsch connector to the ECM. Tighten the screw using a 4-mm [5/32-in] hexagon head wrench.

Torque Value: 3 n·m [27 in-lb]

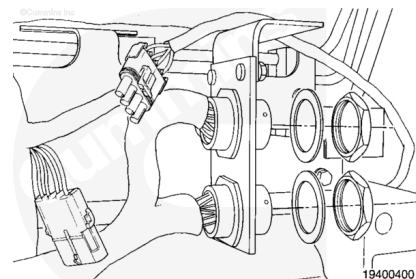


Fasten the harness to the harness support brackets using wire ties.

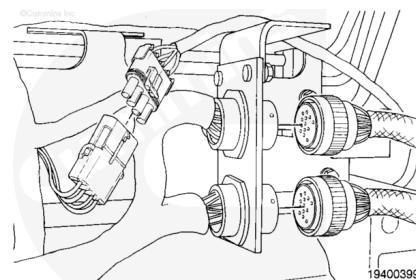


Install the 21- and 31-pin connectors to the harness support brackets.

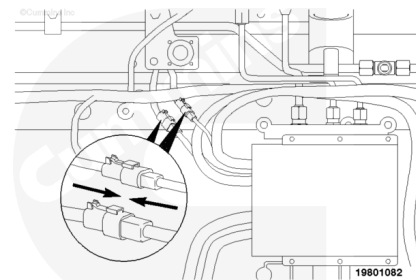
Torque Value: 16 n·m [142 in-lb]



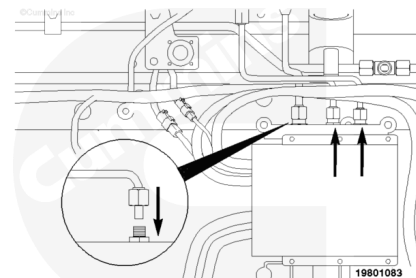
Connect the 21- and 31-pin connectors to the OEM harness.



Connect the 2- and 6-pin Deutsch connectors.



Install the fuel supply tubes to the top of the control valve body. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/19/19-066-024.html>) in the Troubleshooting and Repair Manual, QSK19 Series Engines, Bulletin 3666098 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666098.html>).



QSK23 Engines

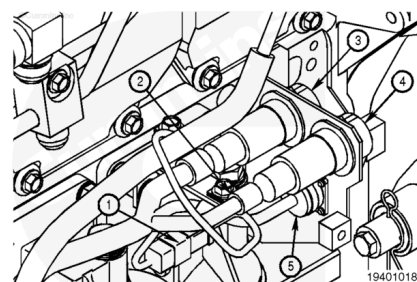
Install the OEM harness.

Install the four capscrews (5) securing the 9-pin connector to the bracket.

Install the connector nuts securing 21- and 31-pin connectors (3, 4) to the bracket.

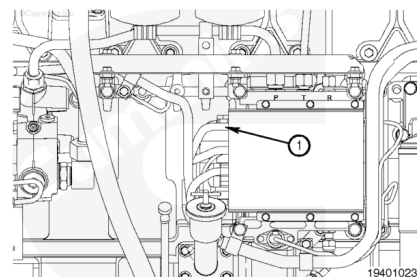
Install the p-clips (2) near the lower right corner of the number 5 cam follower cover.

Connect the datalink (1) connector from the engine harness to the OEM harness.

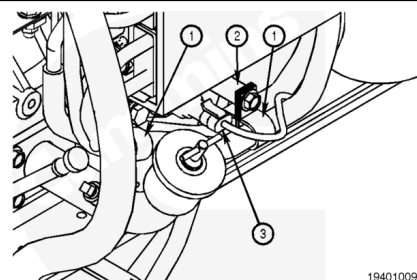


Connect connector B (1) to the ECM.

Install the eight wire ties securing the OEM harness to the engine harness.



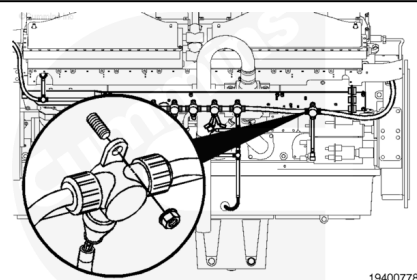
Install the p-clips (1) and bracket (2) near the bottom of the ECM near the ambient air pressure sensor (3).



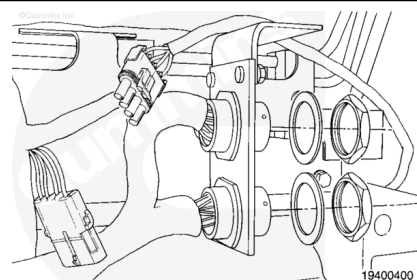
QSK45 and QSK60

Position the OEM interface harness into the three wire clamps.

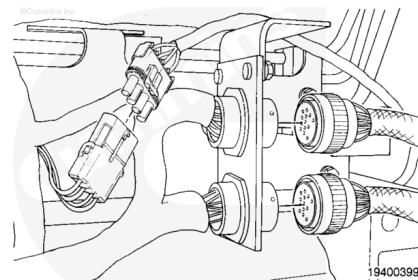
Install the three wire clamps on the engine harness support bracket, using the six attaching nuts to secure the clamps.



Install the OEM interface harness 21- and 31-pin connectors on the engine harness support bracket.

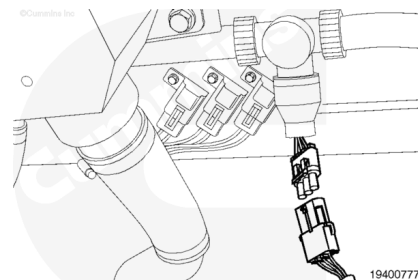


Connect the 21- and 31-pin OEM interface harness connectors to the OEM harness.

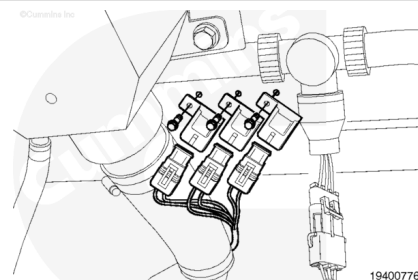


If CENSE™-equipped, connect the OEM interface harness to the CENSE™ harness.

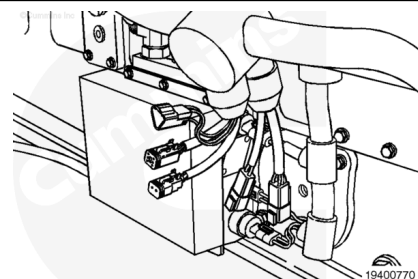
Note : On engines not equipped with CENSE™, the CENSE™ connectors will be connected to each other and must remain that way for proper operation.



Secure the three fuse holders to the engine harness support bracket.

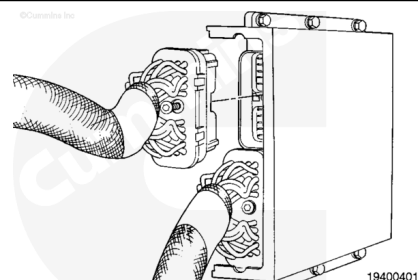


Connect the three datalink connectors to the OEM interface harness.



Connect the OEM interface harness Deutsch connector to the ECM. Tighten the screw using a 4-mm [5/32-inch] hexagon head wrench.

Torque Value: 3 n•m [27 in-lb]



Install the breather box assembly.

- Connect the breather box to the harness support bracket with six bolts (8).

Torque Value: 45 n•m [33 ft-lb]

- Connect the lower breather hose (7) with the hose clamp (6).
- Connect the blowby pressure sensor hose (5) with the hose clamp (4).
- Connect the two hoses (2) on the bottom of the breather box (3) with the two hose clamps (1).

